

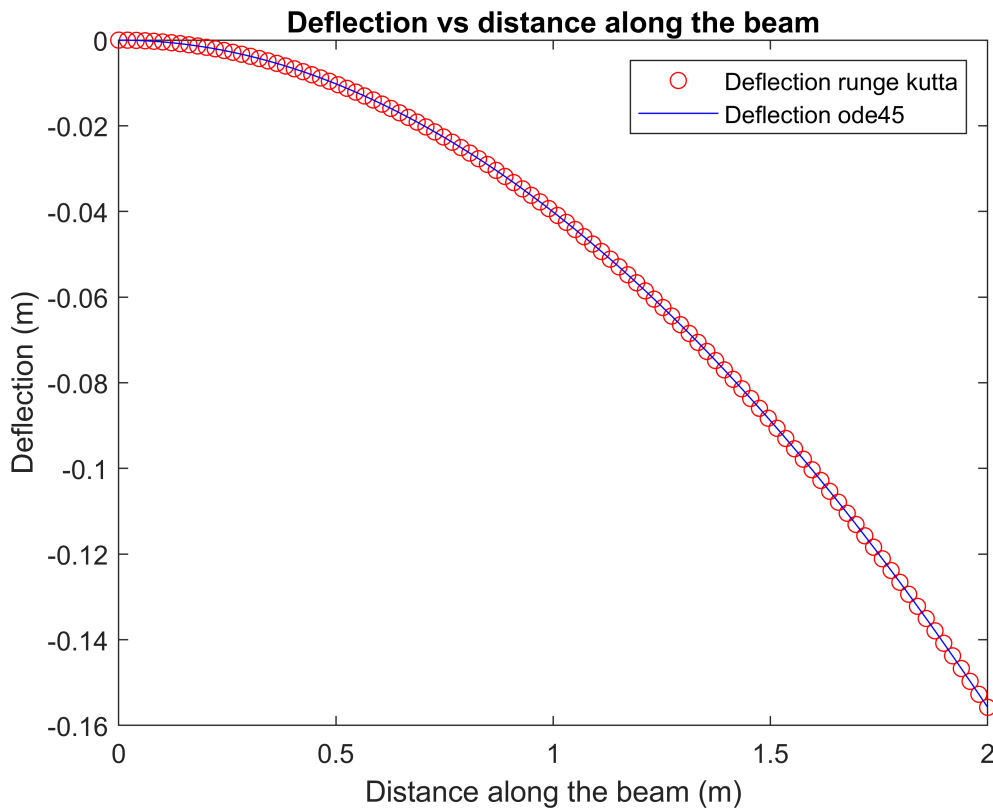
Exercise 6:

```
bc = [0; 0];
t=linspace(0,2);
func=@odesyst;
[a,b] = ode45(func,t,bc);

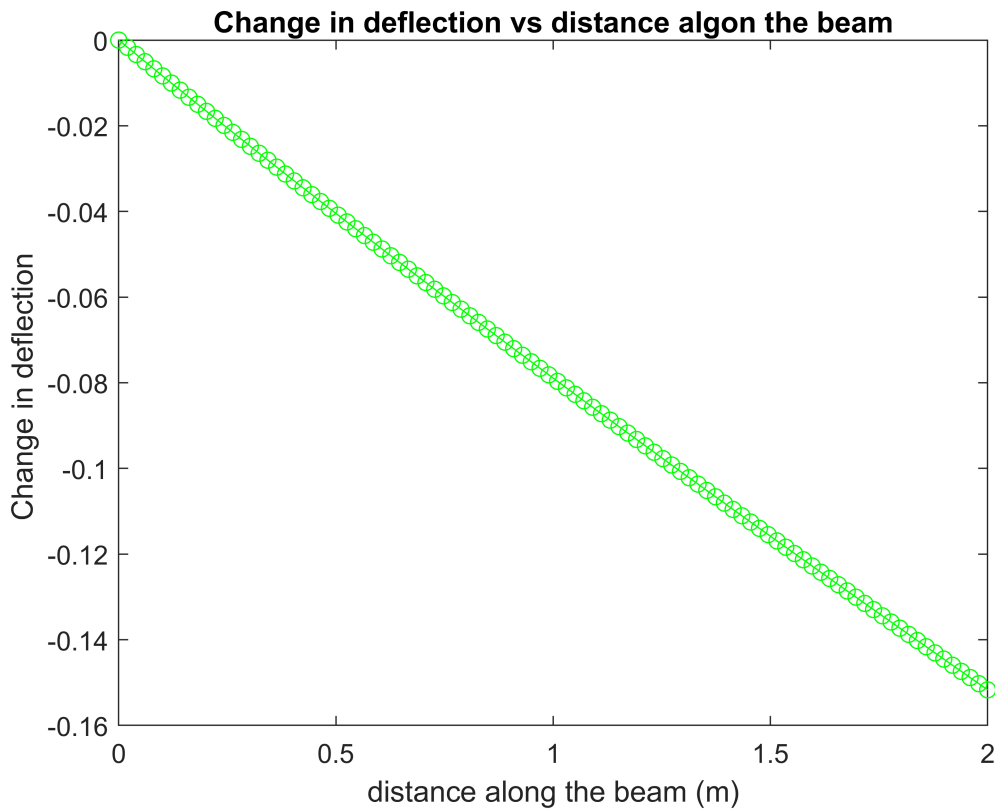
yode45 = b(:,1);
dydxode45 = b(:,2);

[results]=runge_kutta(func,t,bc);
y = results(1,:);
dydx = results(2,:);

plot(t,y,'ro',t,yode45,'b')
title('Deflection vs distance along the beam')
legend('Deflection runge kutta','Deflection ode45')
xlabel('Distance along the beam (m)')
ylabel('Deflection (m)')
```



```
plot(t,dydx,'go',t,dydxode45,'g')
title('Change in deflection vs distance along the beam')
xlabel('distance along the beam (m)')
ylabel('Change in deflection ')
```



```
function [dydx] = odesyst(t,y)
L=2; rhog=100; EI=2400;

dydx = zeros(2,1);

dydx(1) = y(2); % dydx

dydx(2) = (1/EI)*(-rhog*sqrt( (1+(y(2)))^3)*(y(1)*(y(1)-L/2)+(L^2/2)));%d^2y/dx^2

end

function [y] = runge_kutta(df,x,y0)
m = length(y0);
n = length(x);
% y is a matrix of size 'mxn'
y=zeros(m,n);
y(:,1)=y0;
for i=1:n-1
    h = x(i+1)-x(i);
    xi=x(i);
    yi=y(:,i);
    k1 = feval(df,xi,yi);
    k2 = feval(df,xi+0.5*h,yi+0.5*k1*h);
    k3 = feval(df,xi+0.5*h,yi+0.5*k2*h);
```

```
k4 = feval(df,xi+h,yi+k3*h);  
y(:,i+1) = y(:,i) + h*(k1+2*k2+2*k3+k4)/6;  
end  
end
```